

A Structured Review of AI-Based Smart Tutor Systems for Adaptive and Personalized Education

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Abstract

This study buddy system is a kind of AI that acts like an elaborate tutor, aiding students to grasp their lessons more thoroughly and more effectively, all while we are not substantially damaging academic outcomes. This learning platform is the first of its kind, with AI-based explanations and a unique Intelligent Document Processing (IDP) approach for creating automatic assessment and personalised study plan development. This entire research takes a conventional research structure in order to complement a conceptual and an operational framework of an AI tutorial system. The study specifically concentrates on four main sections: first studying materials, followed by AI based explanation processing and summary generation, then automatically generated exam questions after the generated content plus intelligent study planning with focus reinforcement. In conclusion, according to these studies, modular AI systems can allow all learners to learn more effectively with reduced cognitive load, improved assessment outcomes and enduring learning behaviour. In the end, however, the guide provides a systematic approach which will allow organizations to build AI tutoring systems that are scalable and ethical while not losing site of the true needs of students.

Keywords: AI tutoring system, smart education, generative AI, automated assessment, personalized learning, AI study planner

1 Introduction

The education system has undergone a major digital transformation, providing students with access to vast amounts of academic content, including PDFs, lecture slides, video transcripts, and other online learning resources. But even with all that, it can still be hard for many learners to organize and actually understand what they get, because older educational platforms often don't offer enough structured guidance or really personalized academic support. On top of that, differences in learning pace, cognitive ability, and even mental well-being needs can mess with both engagement and academic results. In the last few years, recent progress in Artificial Intelligence (AI), natural language processing, and adaptive learning technologies has changed how digital education systems work. These tools help with automated document handling, smarter summarization, individualized learning paths, and even automated assessment creation. AI-based tutoring systems can turn unstructured academic material into something more usable and interactive, while also trying to match different learner needs. Still, A lot of existing work concentrates on just one piece of the puzzle, like document analysis, text simplification, adaptive learning, or assessment generation, instead of building one connected educational structure. Also, practical barriers—like combining systems smoothly, personalization quality, explainability, and cognitive support—keep intelligent tutoring systems from being implemented well in real settings. So this paper reviews AI-driven educational support systems by looking at their learning impact, their technical strengths and how feasible they are in practice. The goal is to tackle the challenges above through these research questions:

RQ-1: How can raw academic documents in different formats be changed into structured, and context-aware content for an AI tutoring setting?

RQ-2: How can generative AI reduce cognitive load by reshaping complicated academic materials into more approachable and individualized learning formats?

RQ-3: How well can an AI-driven system craft assessments that measure student understanding across multiple cognitive levels and different question kinds?

RQ-4: How can AI-supported educational systems improve academic performance via personalized study plans and mental health-aware learning assistance?

2 Literature Review

Artificial Intelligence (AI) is increasingly becoming an important tool associated with modern edtech showing up in personalised learning, machine grading, and intelligent tutoring systems among other uses. According to more recent studies, actual improvements in document processing, text simplification, automated question generation, and student-adaptive learning systems are in the offing (Kabudi et al., 2021 ; Rodrigues et al., 2025). When you think about work further back in time the emphasis looks more like separate AI tools, assessment systems, text simplification models, tutoring systems and so on. Nonetheless, the newer work seems more inclined toward blended approaches or stitched up strategies or more personalized learning support rather than just stand alone modules or disjointed capabilities (González-Calatayud et al., 2021 ; Tong Ren, 2025).

2.1 Insights From Research Articles

The studies that were reviewed emerged to highlight some significant hurdles that are present in AI-enabled education systems and it seems like researchers are facing the same sort of wall again and again. To begin with, the conversion of messy academic documents into something machines can read is never easy, like, at all. The second part is where you need to keep everything accurate as the text itself is made simpler, and I don't mean summaries, the meaning has to stay intact. The next part that seems the hardest for many is making assessments effectively pedagogy sound (Jaha, 2024; Bahrainian et al., 2024; Zhu et al., 2025) at producing work. The current systems still cannot personalize, think, or implement unrealistically. Most of the failings are a result of limited computing and narrow data sets (Liu et al., 2022; Kasem et al., 2022).

More recent work tends to lean on deep learning, transformer-based models, and sometimes hybrid AI architectures, all to make educational support more useful in practice. When it comes to document analysis, semantic learning approaches show up a lot, and for adaptive simplification plus summarization it leans heavily on transformer models (Mshali van Zyl, 2023; Zhang et al., 2025). Once the focus shifts towards automatic evaluation, a number of researchers combine semantic similarity information, knowledge graphs and machine learning approaches to create as well as grade questions (Zhu et al., 2025; Zou et al., 2022). Neural networks as well as prediction models are typically used in personalizing learning to guide learning paths, in addition to predicting the performance of students (Tong Ren, 2025; Liu et al., 2022).

Put simply, what tends to be a mixture of adaptive simplification systems, cognitive monitoring tools, automated evaluation frameworks and AI-assisted learning planners. Several studies indicate that the results are robust, sometimes exceeding 85% accuracy, and also enhancing learning impact and assessment quality (Bahrainian et al., 2024; Daza et al., 2024). Regardless, most papers seem to end up getting stuck on the same weak points, like dataset diversity, scalability, interpretability, and whether the different AI components actually fit together in practice (Rodrigues et al., 2025; Zhang et al., 2025). All of this suggests the need for a unified tutoring framework, such that document processing, content simplification, assessment generating and personalized studying can combine together as one integrated system for education.

2.2 Findings From Systematic Reviews and Surveys

The studies reviewed make it look like AI techniques can indeed be of help in case of adaptive learning, automated assessment as well as intelligent tutoring systems (Kabudi et al., 2021). Research has shown that although many apps are stuck in experimental settings, AI-based systems provide helpful personalization and automated feedback (Rodrigues et al., 2025) and are successful in enhancing education at scale. Concurrently, progress in automated question generation and text recognition raises ongoing concerns involving explainability, language accessibility, and ethical deployment (Mitroulias Sioutas, 2026; Khan et al., 2025).

The literature overall shows that there has been a substantive advancement on many AI uses in education, but still many papers focus on just one task for one process rather than connected systems which could be termed end-to-end. It is clear what the research gap is: an all-in-one AI tutoring framework that brings together document analysis, content simplification, automated assessment and individual learning support

in one coherent intelligent system. These are not just individual modules that work on their own, and then it's done.

3 Methodology

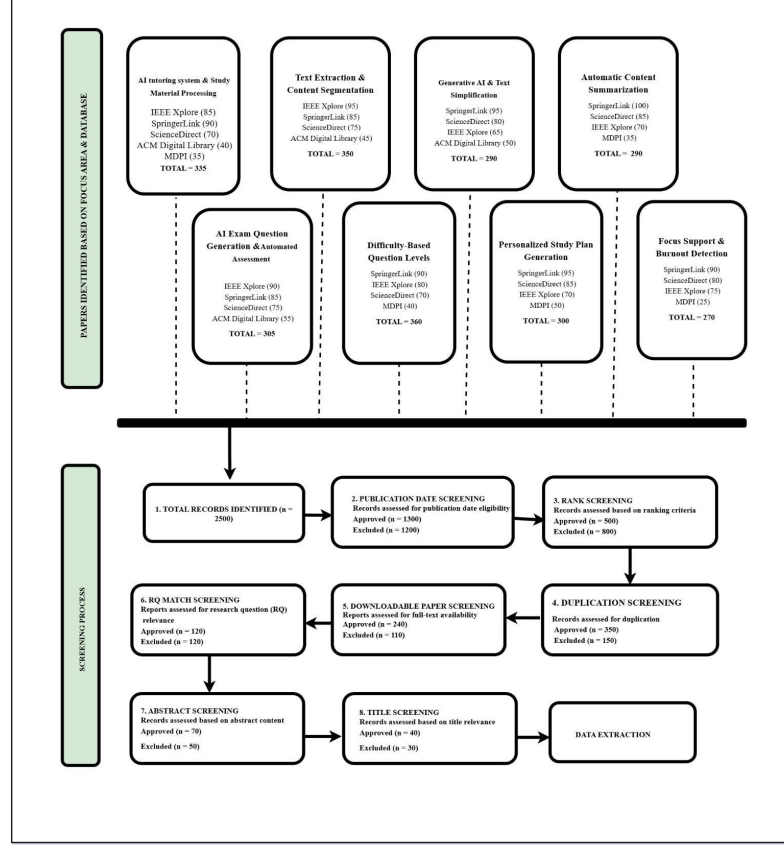


Fig. 1: Systematic screening process for paper selection.

Fig. 1 introduces a systematic process to screen papers. A systematic literature review was conducted by the researchers across the bigger digital libraries such as IEEE Xplore, ScienceDirect, MDPI, ACM Digital Library, and arXiv to collect papers related to AI, NLP, and EdTech. Rephrase

More

Undo They then used Boolean operators with a set of keywords which included smart tutoring systems, text simplification, automated question generation, Explainable AI, burnout detection, and personalised learning. The initial pool of studies for

the present analysis was assembled using the forward and backward citation tracking method. Together these two methods yielded somewhere near 2500 studies that became part of the initial studies.

Subsequently, the review retained only English peer-reviewed journal and conference articles from 2021 to 2026, with an additional emphasis on Q1 journals and A/A*-ranked conferences. Any content that did not fit education AI and that missed explicit AI methods, decided not be peer reviewed, did not validate enough, got removed.

After applying their filters, removing duplicates, checking the full texts and running the relevance screening, they ended up with 40 studies. Subsequently, the paper systematically examined these papers to identify their objectives, approaches, results, emerging trends, and the still unresolved research gaps in AI-based tutoring systems.

4 Results

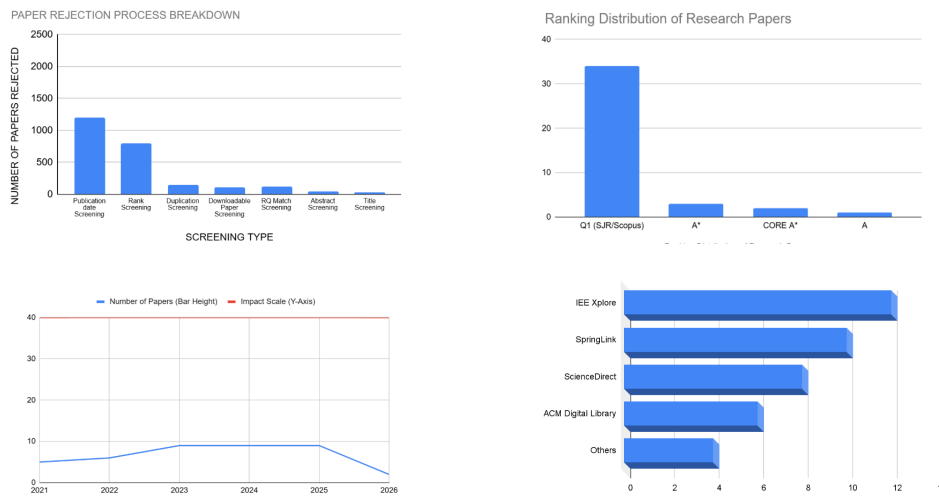


Fig. 2: Overview of paper screening, ranking distribution, publication trends, and data sources

Fig. 2 essentially lays out a complete performance check and comparison of a few research studies, and you can see, more or less, the steps in paper screening, how the ranking shows up, what the yearly publication pattern looks like, and also the contributions broken down by database. The results indicate that a great majority of the studies (like most of them) are coupled together in top-class Q1 venues. The other major sources are IEEE Xplore and SpringerLink. The analysis appears to reflect a kind of systemized procedure as well, involving the employment of appropriate, quality, and reliable literature in steps..

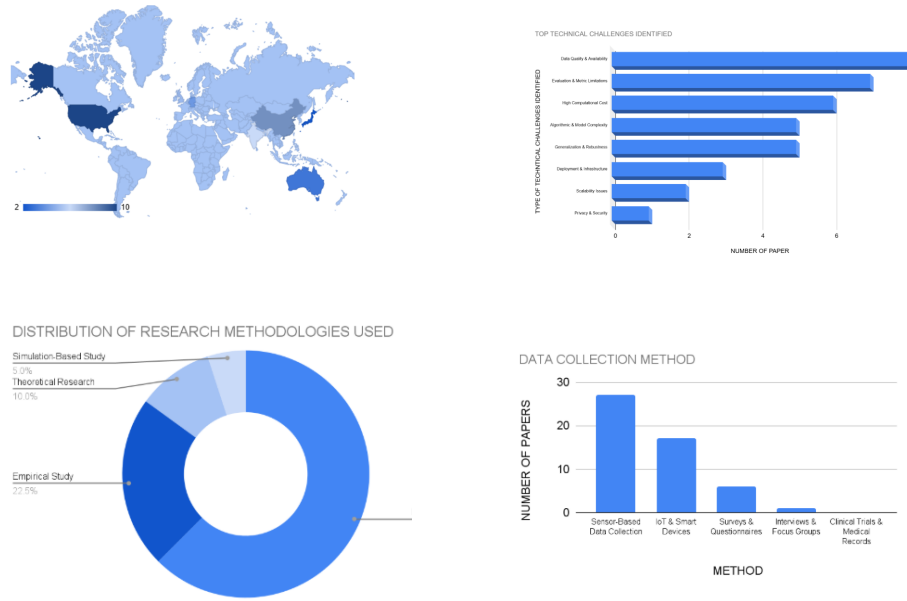


Fig. 3: Summary of global research distribution, focus areas, methodologies, and data collection methods

A detailed breakdown (Fig. 3) of research by geographic region, focus areas, methodologies, and data collection approaches is presented in, showing a high concentration of studies in certain regions and highlighting dominant focus areas such as machine learning and adaptive systems, with literature reviews and experimental studies being the most common methodologies and sensor-based and IoT data collection approaches being the most common data collection approaches, respectively.

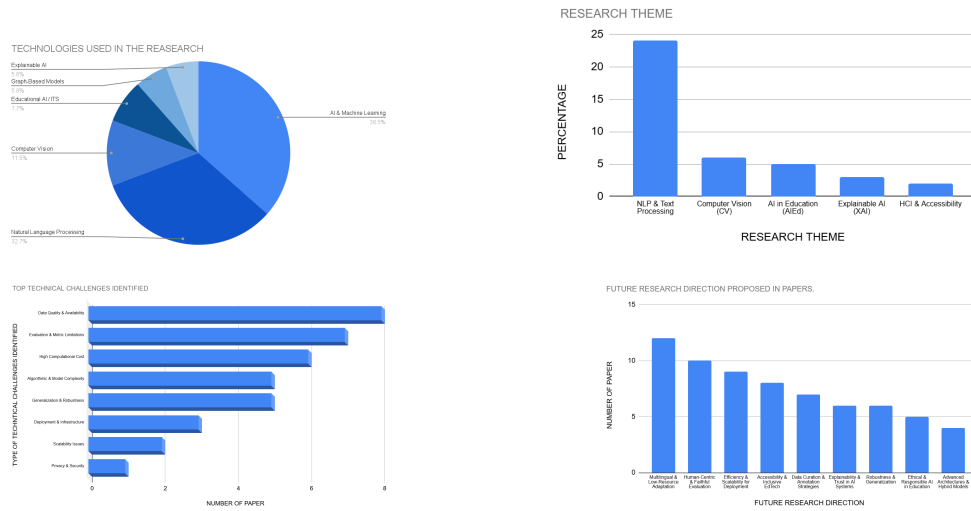


Fig. 4: Overview of technologies used, research themes, key challenges, and future directions

Fig. 4 summarizes the technological areas, research topics, key challenges, and future research directions identified from the studies selected. The majority of technologies are in the areas of machine learning and natural language processing, and AI in education is the main research topic, with data quality and model limitations being the main challenges, and personalization, explainable AI, and adaptive learning systems being the emerging directions.

Table 1: Comparison of related works in AI-enabled educational frameworks

Work	Scope	Key Characteristics	Results
This work	Integrated Tutoring Framework	Document processing, simplification, automated assessment, and planning.	Proposes the “AI Study Buddy” to bridge gaps between isolated domains.
Kabudi et al. (2021)	AI-enabled adaptive learning systems	Focus on personalization and implementation frameworks.	Implementations remain experimental; faces challenges in prescriptive analytics.
González-Calatayud et al. (2021)	AI for student assessment	Automated grading and formative assessment focus.	Pedagogical foundations are weak; lacks teacher training and integration.
Mitroulias & Sioutas (2026)	Automated MCQ generation	Bibliometric analysis of research activity and clusters.	Identified growth in US contributions and emerging ethical/domain clusters.
Rodrigues et al. (2025)	AI in engineering education tutoring	Integration of personalization and real-time feedback.	Personalization exists, but dual-user (student/instructor) support is lacking.
Khan et al. (2025)	Machine learning for text recognition	Evolution of models and character recognition.	Technological growth exists, but multilingual and degraded document support is low.

The Table 1 compares the existing AI educational frameworks against this proposed AI Study Buddy system, in terms of scope, feature sets, and outcomes. Most of the previously worked-on jobs mostly focus on a single aspect such as learning personalization, automated evaluation, etc. The system proposed here seems to be more integrated where the actions of document processing and assessment generation get integrated with the personalized study plan to build a flow of connection between them.

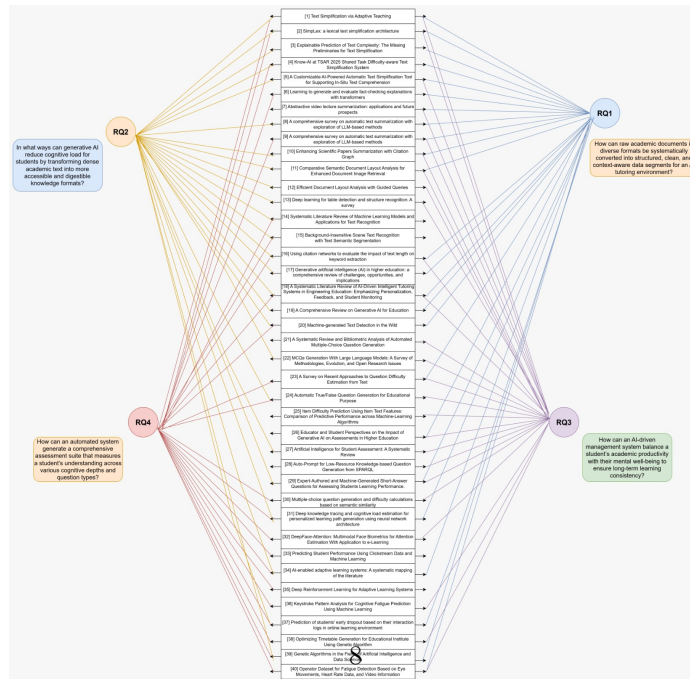


Fig. 5: Breakdown of the impact made by the number of research papers

Fig. 5 shows RQ mapping the studies link to the four research questions through themes such as text simplification, assessment generation and student support bringing out the interdisciplinary nature of AI driven educational systems. While this emphasizes the research coverage that we understand is quite clear at this stage, there is still a gap.

5 Conclusion

Researchers who developed the AI Study Buddy reviewed 40 studies to determine whether this smart tutoring system was an idea and found that the four major components are indeed quite effective: The AI Study Buddy can take unstructured school work and format it into something that makes sense, it can model text to make it more easily understandable, it can model test questions appropriate for each student, and it can attempt to predict student performance as well as their likelihood to stay engaged in school work. The AI Study Buddy is a computer-based learning tool that its developers believe holds promise as a potential tutor. The AI Study Buddy has four major components that they found are very effective: It can take unstructured school work and format it into something that makes sense, it can model text to make it more easily understandable, it can model test questions appropriate for each student, and it can attempt to predict student performance as well as their likelihood to stay engaged in school work. It is a computer-based learning tool that its developers believe holds promise as a potential tutor.

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